[LOGO AREA]

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CYBERSECURITY | VAPT

SECURITY ASSESSMENT REPORT

TryHackMe – What is Networking?

REPORT ID	AZ-EH-2026-004
VERSION	1.0
ASSESSMENT DATE	17/10/2026
REPORT DATE	22/10/2026
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CLASSIFICATION	Public

Overview

This room introduces the basics of computer networking and how devices communicate with each other. It covers the Internet, IP and MAC addresses, and the `ping` command.

Networking is something I need to understand properly for penetration testing. Before scanning a target or investigating a service, I need to know how devices identify each other and how data travels between them.

1. What Is Networking?

A network is a group of devices connected so they can communicate and share information.

The Internet is a large network of interconnected networks. When I access a website, my device communicates with a server over this infrastructure.

The room also introduces the World Wide Web, which was invented by Tim Berners-Lee. The Web is a service that operates over the Internet, rather than being the Internet itself.

2. Identifying Devices on a Network

Devices need ways to identify themselves and communicate with other devices. Two important concepts are IP addresses and MAC addresses.

IP address

An IP address is used to identify a network interface for communication over an IP network.

An IPv4 address looks like this:

```
192.168.1.10
```

It contains four numbers separated by dots. Each section is called an octet because it represents eight bits.

IP addresses help packets reach the correct destination across networks.

MAC address

MAC stands for Media Access Control. A MAC address is used at the data-link layer to identify a network interface on a local network.

IP and MAC addresses serve different purposes. IP addresses help with communication between networks, while MAC addresses are used for local-link delivery.

The room includes a practical exercise involving changing a device's MAC address to access a lab website.

3. Ping and ICMP

`ping` is a command-line utility used to test whether a destination responds to network probes. It normally uses ICMP Echo Request and Echo Reply messages.

For example:

```
ping 10.10.10.10
```

This sends ping requests to the specified IP address.

If replies come back, it indicates that the destination is reachable through the tested path at that time. If there are no replies, it doesn't necessarily mean the device is offline; a firewall or network configuration might block ICMP traffic.

The room also uses `ping` to interact with a lab target and retrieve a flag.

Conclusion

This room covered some of the basic concepts behind how devices communicate. IP addresses, MAC addresses, and ICMP might seem simple at first, but they're important foundations for understanding network scanning and troubleshooting.

As I continue learning penetration testing, I'll need to build on these concepts to understand how hosts are discovered, how services communicate, and how network traffic can be analyzed.

